

Previous examination subject for FoEC

Note: Course content and examination subject can differ

Exam , June 2006

T1. 1.5p. What is the circuit for the NAND CMOS gate? Describe the operation of the circuit by means of the truth table and the states of all transistors (*on* or *off*).

T2. 1.5p. The operational amplifier: symbol; terminals; operation equation; properties of the ideal op amp.

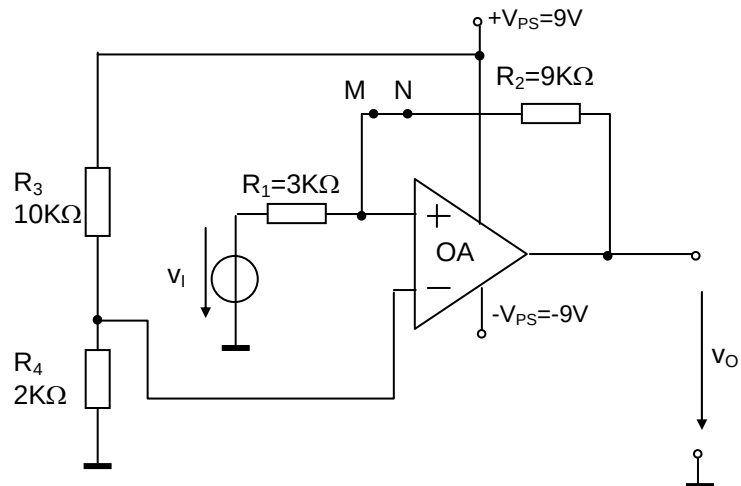
E1. Assume OA – ideal, rail-to-rail.

0.75 p a) Draw, qualitatively, the VTC $v_O(v_I)$ of the circuit. What is the function of the circuit?

0.75 p b) Find the expressions and values of the following parameters from VTC: V_{OH} ; V_{OL} ; $V_{Th,H}$; $V_{Th,L}$. Re-draw the VTC $v_O(v_I)$ according to the numerical values obtained.

0.75 p c) Assume $v_I(t)$ is a triangular wave, with no dc component, of 8V amplitude. Plot $v_I(t)$ and $v_O(t)$. How does $v_O(t)$ change if the amplitude of v_I reduces to 4V?

0.75 p d) How does VTC change if in the circuit appear an interruption between points M and N and what is the function of the new circuit?

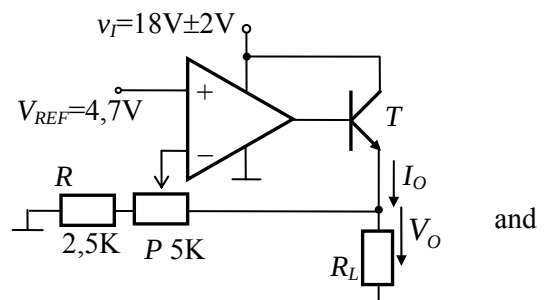


E2.

a) 1p. What is the application of the circuit? What are the expressions and values of V_{Omax} , V_{Omin} ?

b) 0.5p. How can be modified the circuit so that I_O will be limited to $I_{Omax}=750mA$?

c) 0.5p. Consider the circuit in the previous question ($I_{Omax}=750mA$) the tap of the potentiometer on the middle position. What is the maximum power dissipated by the transistor T in case of an output shortcircuit?



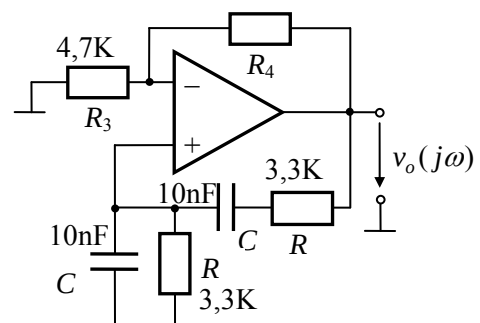
E3.

a) 0,5p. What is the application of the circuit? What are the expression and value of the oscillation frequency?

b) 0,5p. What should be the value of the R_4 resistor to accomplish the oscillation criterion?

c) 0.5p. Assume a value of 6V for the output voltage. What does the output voltage and the voltages at the inverting and noninverting input look like?

d) 0,5p. Complete the circuit in order to obtain the automatic control of the amplitude.



T1. 1.5p. What is the circuit for the NOR CMOS gate? Describe the operation of the circuit by means of the truth table and the states of all transistors (on or off).

T2. 1.5p. The operational amplifier: symbol; terminals; operation equation; properties of the ideal op amp.

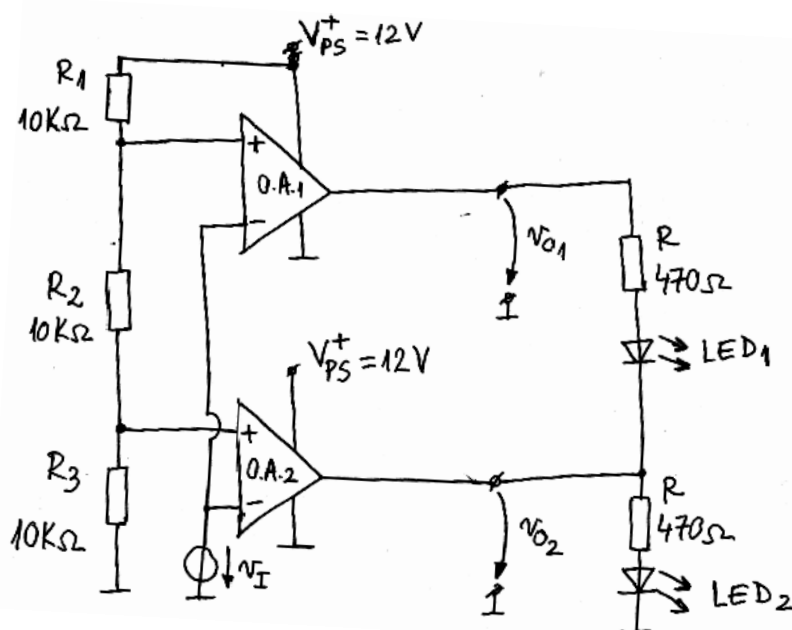
E1 OA₁, OA₂ – ideal

1p a) What is the expression $v_{O1}(v_I)$ for $v_I \in [0V; 10V]$? Plot the VTC $v_{O1}(v_I)$. What is the application of the circuit with the output v_{O1} ?

1p b) What is the expression $v_{O2}(v_I)$ for $v_I \in [0V; 10V]$? Plot the VTC $v_{O2}(v_I)$. What is the application of the circuit with the output v_{O2} ?

1p c) Fill in the table of states (on/off) of the two LEDs below:

v_I range	State of LED ₁ (on/off)	State of LED ₂ (on/off)

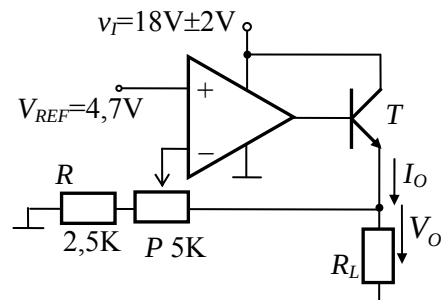


E2.

a) **1p.** What is the application of the circuit? What are the expressions and values of V_{Omax} , V_{Omin} ?

b) **0.5p.** How can be modified the circuit so that I_O will be limited to $I_{Omax}=750mA$?

c) **0.5p.** Consider the circuit in the previous question ($I_{Omax}=750mA$) and the tap of the potentiometer on the middle position. What is the maximum power dissipated by the transistor T in case of an output shortcircuit?

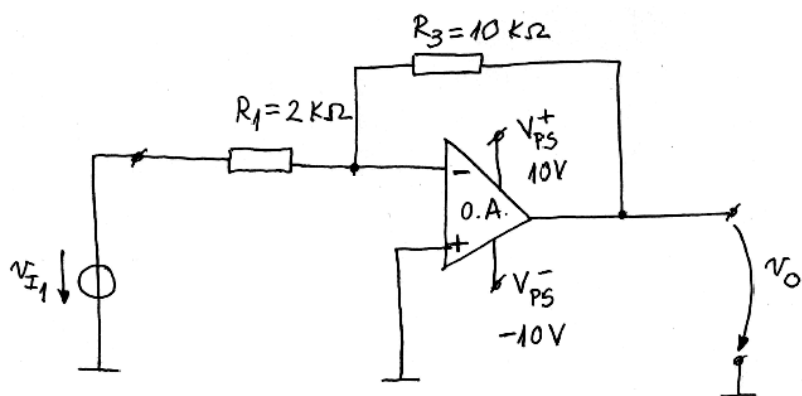


E3. OA – ideal

1p a) Find the expression $v_O(v_{I1})$ assuming the range of v_{I1} small enough to keep OA in the active region. What is the application of the circuit?

0.5p b) Plot $v_O(t)$ for $v_{I1}(t)=3\sin\omega_0 t$ [V]

0.5p c) What are the values of: the input resistance R_{i1} seen by v_{I1} and the output resistance R_o of the amplifier?

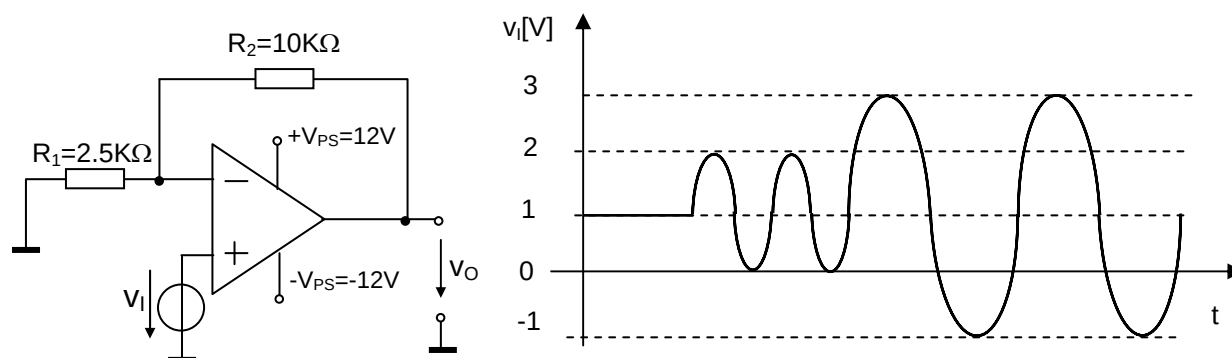


Exam "Fundamentals of Electronic Circuits" june 2007

T1. 2,5p. Op-amp voltage regulators ($V_O > V_{REF}$; $V_O < V_{REF}$; adjustable V_O , $V_{Omax} > V_{REF}$ $V_{Omin} < V_{REF}$; for each circuit: the circuit and expression of V_O).

T2. 1,5p. Half-wave rectifier with capacitive filter: circuit, waveforms, the effect of the filtering capacitance on the output voltage shape – from the qualitative point of view.

P1. 2p



Assume a rail-to-rail op amp.

0.5 p a) What is the expression and the value of the gain and how does the VTC, $v_O(v_I)$ look like considering $v_I \in [-5V; 5V]$? What is the v_I range for that the amplifier remains in its active region?

0.5 p b) What are the values of the input and output resistances. What is the application of the circuit?

0.5 p c) How does the $v_O(t)$ look like for $v_I(t)$ in the above figure?

0.5 p d) Where another source v_{II} should be connected to obtain $v_O = 5v_I - 4v_{II}$?

P2. 2p

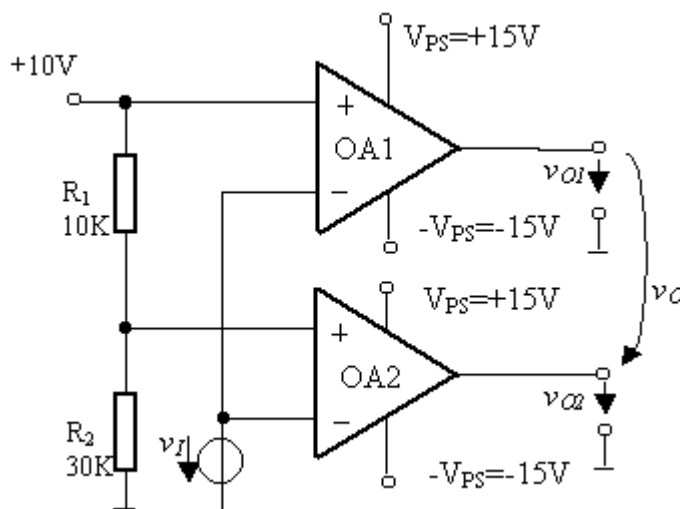
Assume OA_1 , OA_2 ideal.

a) 0.5p For $v_I \in [-12V; 12V]$, find the expression of $v_{O1}(v_I)$ and plot the VTC $v_{O1}(v_I)$. What is the application of the circuit assuming v_{O1} as output?

b) 0.5p For $v_I \in [-12V; 12V]$, find the expression of $v_{O2}(v_I)$ and plot the VTC $v_{O2}(v_I)$. What is the application of the circuit assuming v_{O2} as output?

c) 0.5p Plot the VTC $v_O(v_I)$ for $v_I \in [-12V; 12V]$.

d) 0.5p Assuming $v_I(t) = 11 \sin \omega t$ [V], plot $v_I(t)$ and $v_O(t)$. What is the application of the circuit?



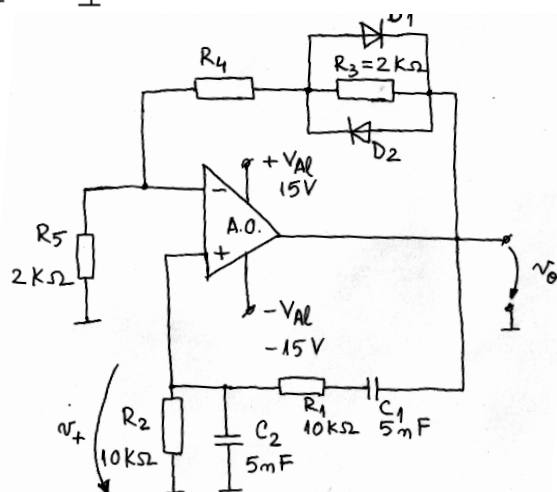
P3. 2p

a) 0,5p How do the $v_o(t)$ and $v_+(t)$ signals look like, qualitative, in permanent regime?

b) 0,5p Compute the frequency of the $v_o(t)$ output signal.

c) 0,5p Size R_4 such that the circuit will sustain the oscillations in steady-state regime. Consider that in conduction, the equivalent resistances of D_1 and D_2 diodes are $r_{D1} = r_{D2} = 0,5K\Omega$. Verify the chosen value for the condition of starting-up the oscillation in transient regime.

d) 0,5p How does the $v_o(t)$ signal shape modifies in permanent regime if the D_2 diode connection is omitted in the circuit?



Exam “Fundamentals of Electronic Circuits” sept 2007

T1. 2p. Inverting op-amp comparator with hysteresis (with positive feedback): circuit, VTC, waveforms to illustrate the operation.

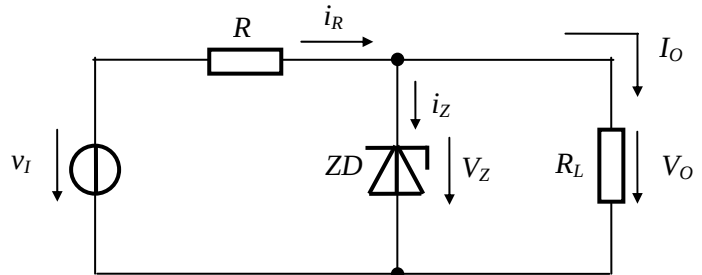
T2. 2p. Rectangular and triangular signal generator - Astable multivibrator with an integrator and a PF comparator (circuit; waveforms of the rectangular and triangular voltages; extremes values of the voltage across the capacitor; the deduction of the oscillation period).

P1. 2p. For the circuit we know the range of $v_I \in [13; 16] \text{ V}$, the type of the Zener diode: PL10Z, the range of $R_L \in [350 \Omega; \infty]$ and $R = 70 \Omega$

a) 0.5p. What is the value of the output voltage?

b) 1p. What are the minimum and maximum values of the i_Z current?

c) 0.5p. What is the maximum power dissipated by the Zener diode?



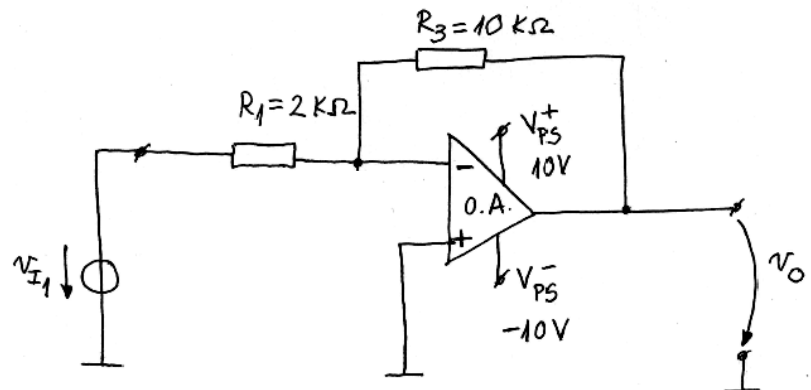
P2. 2p

OA – ideal

1p a) Find the expression $v_O(v_{I1})$ assuming the range of v_{I1} small enough to keep OA in the active region. What is the application of the circuit?

0.5p b) Draw the voltage transfer characteristic of the circuit.

0.5p c). Plot $v_O(t)$ for $v_{I1}(t) = 3 \sin \omega_0 t \text{ [V]}$



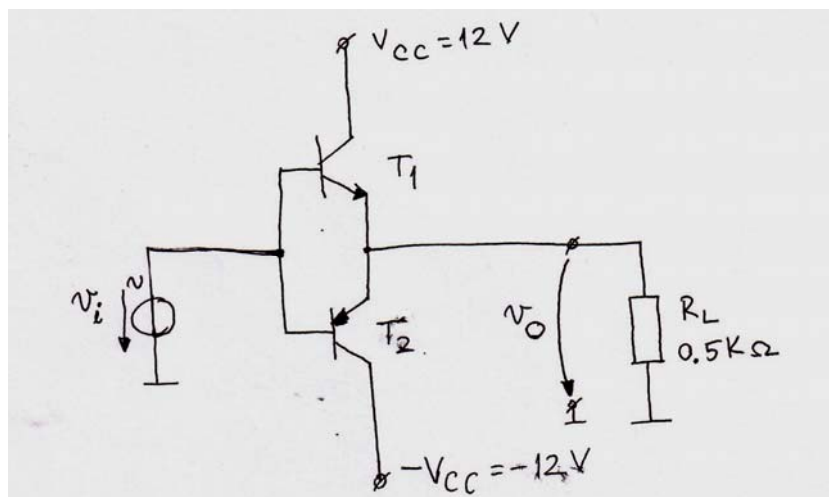
P3. 2p

For T_1 and T_2 , $|V_{BE,on}| = 0.6 \text{ V}$; $\beta = 20$.

a) 1p What is the expression $v_O(v_i)$ for $v_i \in [-10 \text{ V}; 10 \text{ V}]$? Plot the VTC $v_O(v_i)$ for this range of values of v_i .

b) 0.5p To what class of output stages does this circuit belong? What is the theoretical value of the maximum efficiency for this class?

c) 0.75p Assuming that, for $v_i(t) = 10 \sin \omega t \text{ [V]}$, $v_O(t)$ can be considered a sinewave, compute the average power transmitted to R_L .



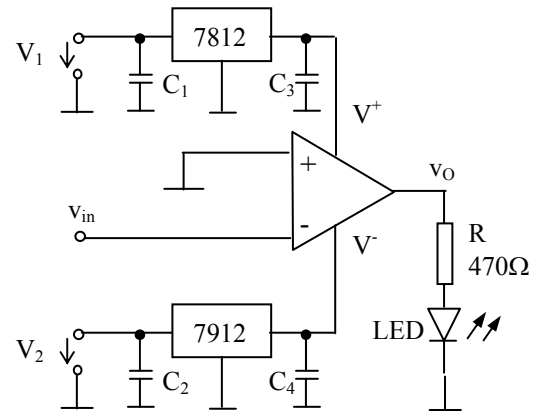
Exam “Fundamentals of Electronic Circuits” sept 2007

T1. 2p. Logic inverter: circuit containing R and a controlled switch, truth table that include the state of the switch; critical analysis, solutions to eliminate the disadvantages including truth tables with the states of the switches, CMOS implementation.

T2. 2p. Op amp and WIEN bridge oscillator (circuit; relations of the oscillation frequency and of the gain of the basic amplifier for both cases: different values of the bridge elements, equal values of the bridge elements).

P1. 3p

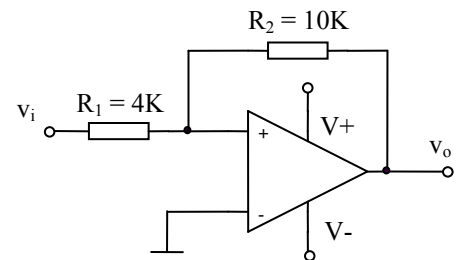
- What are the roles of 7812 and 7912 ICs? What are the values of V^+ and V^- ?
- How does the voltage transfer characteristic $v_O(v_{in})$ look like?
- What is the state of the LED (lighting or not lighting) for each of the following values of v_{in} : +5V; -5V; +0.5V and 0V?
- What is the role of R resistor?



P2. 3p

The supply voltages for this voltage comparator are $V^+ = +15V$ and $V^- = -15V$.

- What are the values of the threshold voltages?
- How does the voltage transfer characteristic $v_O(v_I)$ look like? Show numerical values on the axes.
- Show the waveforms of $v_O(t)$ and $v_I(t)$ for $v_I(t)$ sine wave with 10V amplitude.
- Show the waveforms of $v_O(t)$ and $v_I(t)$ for $v_I(t)$ sine wave with 5V amplitude.



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T2. 2p

Op amp and WIEN bridge oscillator (circuit; relations of the oscillation frequency and of the gain of the basic amplifier for both cases: different values of the bridge elements, equal values of the bridge elements).

P1. 2p

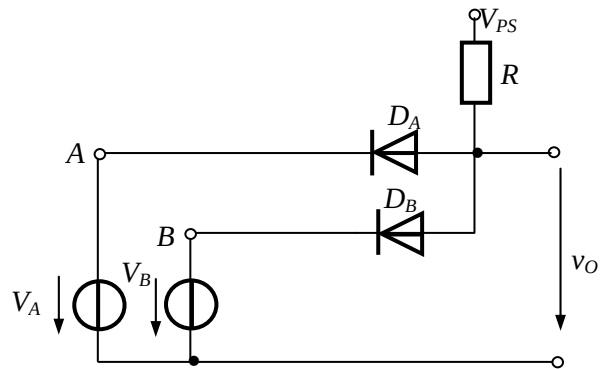
Consider ideal diodes.

0.5p a) Find the expression of the output voltage $v_O(t)$.

0.75p b) Draw $v_O(t)$ for $V_{A1} = 10V$, $v_A(t)$ – triangular voltage with 8V amplitude and $v_B(t)$ – rectangular voltage with 4V amplitude; both $v_A(t)$ and $v_B(t)$ have the same frequency.

0.75p c) Assume the logic convention: 0V – “0”; 10V – “1”

and $v_A, v_B \in \{0V, 10V\}$ draw the electrical operating table (with the state of the diodes – on or off) and truth table of the circuit. What is the logic function of the circuit?



P2. 2p

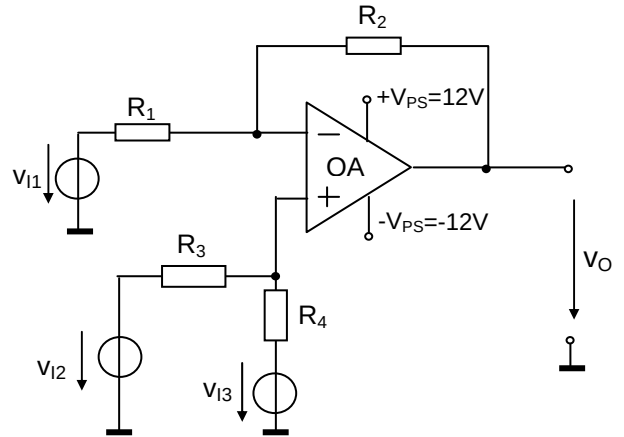
Assume OA – ideal

0.75p a) What is the expression of $v_O(v_{I1}, v_{I2}, v_{I3})$ assuming OA in the active region for $R_1=R_2=R_3=R_4=15K\Omega$?

0.5p b) Now consider $v_{I3}=0$. Find the relationship between R_1, R_2, R_3, R_4 in order to obtain the expression: $v_O=3(v_{I2}-v_{I1})$? For what range of values of $(v_{I2}-v_{I1})$ does OA work in the active region?

0.25p c) Deduce the expressions for input resistances considering the input only v_{I1} (with v_{I2} set to zero), respectively only v_{I2} (with v_{I1} set to zero), and the output resistance of the circuit.

0.5p c) If $v_{I3}=0$ and $v_O=3(v_{I2}-v_{I1})$ plot $v_O(t)$ assuming $v_{I1}(t)=2\sin\omega_0 t[V]+1V$ si $v_{I2}(t)=-3\sin\omega_0 t[V]+4V$. What is the application of the circuit?



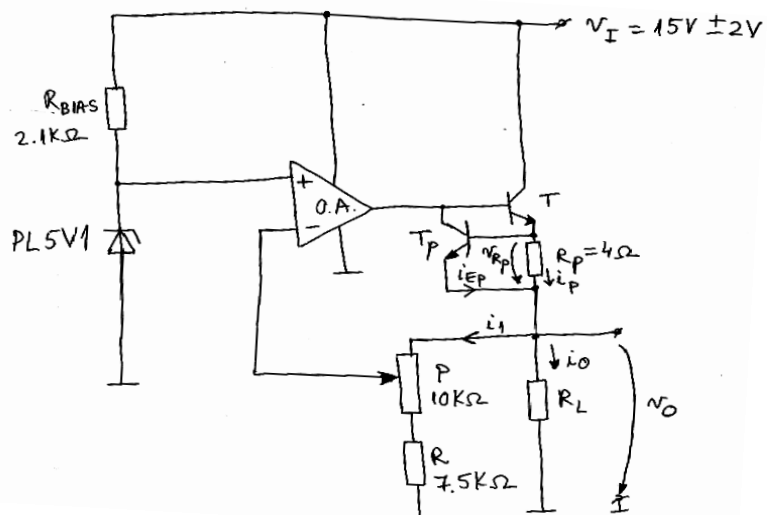
P3. 2p

Consider the O.A. – ideal.

0.75p a) Find the expression and range of values in which V_O can be adjusted. What are the roles of: the Zener diode; O.A.; T; P and R? Assume again that R_L is large enough to neglect v_{Rp} and to maintain T_p – off.

0.5p b) What components in the circuit compose the protection circuit? Explain the protection mechanism.

0.75p c) What is the maximum value of the collector current through T? Assume the base currents of T and T_p are negligible. Assuming the cursor of P in the middle, compute the maximum power dissipation on T for $R_L=0$.

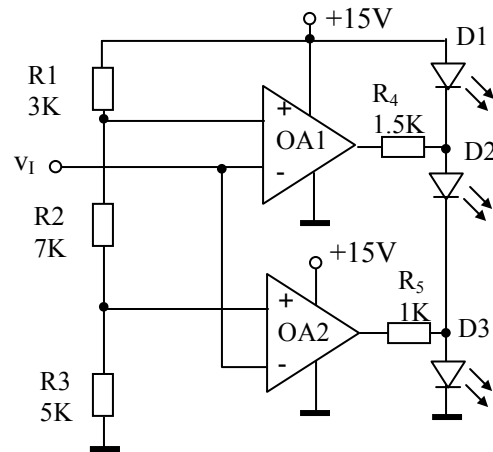


P1. 2p

a) 0.5p What are the applications of the two op-amps? Find the expressions and values of the threshold voltages for OA1 and OA2.

b) 1p What are the states of the diodes D_1 , D_2 , D_3 (on, off) for v_i in $(0; +15V)$?

c) 0.5p Compute the value of the current through D_3 , for D_3 – on, considering that the voltage drop across D_3 – on is 2V.

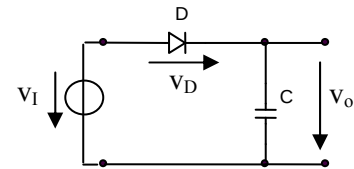
**P2. 2p**

Consider D – ideal, $v_C(0) = 0V$

a) 0.75p For $v_i(t) = 5\sin\omega t$ [V], plot: $v_O(t)$; $v_D(t)$. Specify the state of the diode (on, off) on the $v_O(t)$ plot. What is the application of the circuit ?

b) 0.5p Redraw $v_O(t)$ and $v_D(t)$ for $v_i(t) = 5\sin\omega t$ [V], assuming the constant voltage drop model for D ($V_{D,on} = 0.7V$) and $v_C(0) = 2$.

c) 0.75p Connect a load resistance R_L at the output of the circuit and consider $C = 100\mu F$, $v_C(0) = 0V$, $v_i(t) = 5\sin\omega t$ [V], $\omega = 100\pi$. Find the value of R_L so that the output voltage ripple Δv_O is smaller than 0.5V.

**P3. 2p**

$R = 1K\Omega$, $R_L = 15\Omega$.

a) 0.5p What is the operating class of the power amplifier? What are the roles of: D_1 , D_2 ; R_1 , R_2 ? What is the value of the maximum theoretical average efficiency?

b) 0.5p Deduce and plot the VTC $v_O(v_i)$ of the circuit considering $v_i \in [-10V; 10V]$.

c) 0.5p How does the waveforms $v_O(t)$; $i_{C2}(t)$; $v_{CE2}(t)$ look like, for $v_i(t) = 5\sin\omega t$ [V]?

d) 0.5p Compute: average power in the load R_L ; average power dissipated by the power supplies; average efficiency for $v_i(t) = 5\sin\omega t$ [V].

